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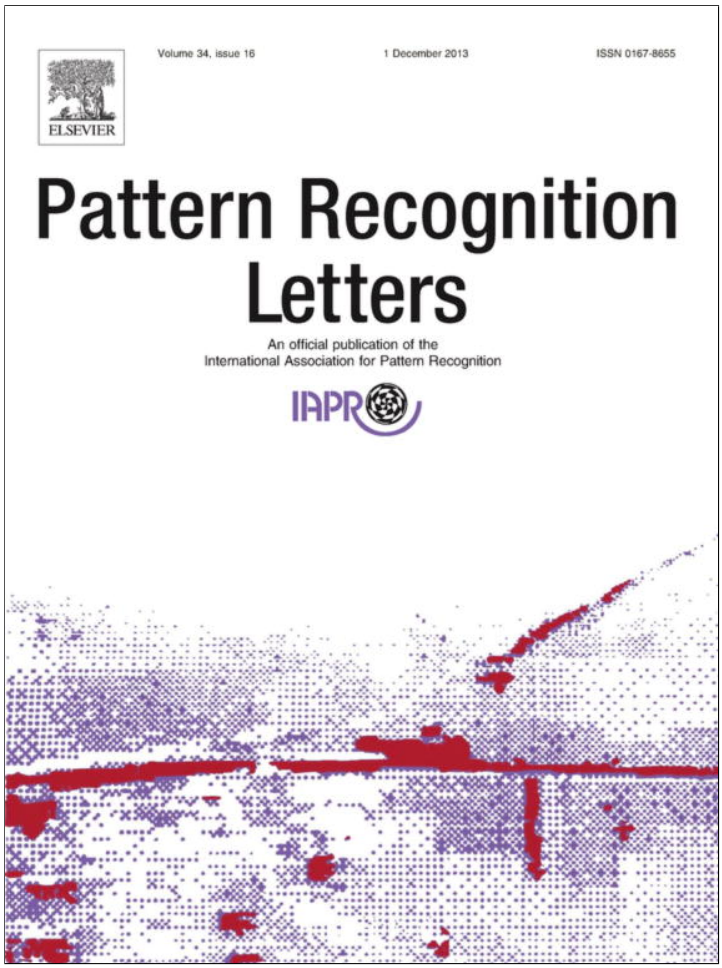

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Fusing cluster-centric feature similarities for face recognition in video sequences

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ABSTRACT

The emergence of video has presented new challenges to the problem of face recognition. Most of the existing methods are focused towards the use of either representative exemplars or image sets to summarize videos. There is little work as to how they can be combined effectively to harness their individual strengths. In this paper, we investigate a new dual-feature approach to face recognition in video sequences that unifies feature similarities derived within local appearance-based clusters. Relevant similarity matching involving exemplar points and cluster subspaces are comprehensively modeled within a Bayesian *maximum-a posteriori* (MAP) classification framework. An extensive performance evaluation of the proposed method on three face video datasets have demonstrated promising results.

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1. Introduction

The emergence of video has presented new challenges to the problem of face recognition, particularly in the abundance of appearance variations, complex head poses and other external conditions. The accessibility of imaging devices and extensive array of applications in consumer products, biometrics and surveillance has rendered automatic face recognition an increasingly important task. Recent surveys on face recognition (Zhao et al., 2003; Li and Jain, 2011) and the well-known Face Recognition Grand Challenge (FRGC) (Phillips et al., 2005) are evident of the significant attention by the computer vision and pattern recognition community to address this problem. Current literature for video-based face recognition can be loosely divided into *still-face-to-video* matching and *video-to-video* matching (Poh et al., 2010).

The notion of *video-to-video* matching typically exemplifies a recognition problem over image sets, which can be in the form of temporally ordered or unordered sets, and consisting of entire sequences or only selected images. Image sets derived from both training and test videos are usually represented by subspaces or manifold models learned from the original vectors. Classification is then accomplished by computation of subspace distances or similarity measures among these learned subspaces. Parametric model-based approaches use Kullback–Leibler Divergence (KLD) metric (Shakhnarovich et al., 2002; Arandjelovic et al., 2005) to evaluate similarities between estimated densities. With difficulty of parameter estimation under limited data, these methods tend to fail when training and test sets are not strongly correlated statistically. Methods based on *principal angles* have a long standing interest due to its simplicity. Mutual Subspace Method (MSM) (Yamaguchi et al., 1998), the seminal work on principal angles for recognition uses the cosine of the smallest principal angle, or largest canonical correlation as a similarity measure. Further improved methods that use principal angles – Constrained Mutual Subspace Method (CMSM) (Fukui and Yamaguchi, 2003), Kernel Principal Angles (KPA) (Wolf and Shashua, 2003) have also demonstrated good results in face recognition. Later on, discriminant methods that maximize the discriminatory power between within-class and between-class sets were also proposed – Discriminative Canonical Correlations (DCC) (Kim et al., 2007), and more recently, a graph embedding discriminant analysis based on Grassmannian manifolds (Harandi et al., 2011).

Lesser often mentioned are exemplar-based representations for face recognition in video, which have also attracted increasing attention in the literature. Exemplar-based representations deal with the large array of face images in training videos by selecting a significantly smaller number of appearance-specific representative face images or *exemplars* to summarize each subject class. In common configurations, clusters or patches are extracted from the data manifold and sample means are chosen as exemplars. Although naturally regarded as a *still-image-to-video* matching approach, it usually requires a fusion of matching scores between exemplars (training set) and multiple query frames (test set) in order to realize a full *video-to-video* matching. Numerous approaches to exemplar selection using radial basis function network (Krüeger and Zhou,

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2002; Zhou et al., 2003), clustering by k-means (Hadid and Peitikäinen, 2004; Fan et al., 2005), Hierarchical Agglomerative Clustering (HAC) (Fan and Yeung, 2006; See and Ahmad Fauzi, 2011) have been proposed. Wang et al. (2008) introduced Maximal Linear Patch (MLP) which partitions data on the manifold into local linear models. More recently, Spatio-Temporal Hierarchical Aggomerative Clustering (STHAC) (See and Eswaran, 2011) was proposed to leverage the use of both spatial and temporal information for the selection of exemplars from ordered images. Relatively good results have been reported for exemplar-based methods although this setup is essentially sensitive towards the effectiveness of clustering appearance variations and the number of exemplars chosen.

Despite claims of superiority of *video-to-video* methods in literature, an analysis from a recent evaluation of video-to-video face verification (Poh et al., 2010) offers an observation that performance of both *still-image-to-video* and *video-to-video* matching are not significantly different and success depends on a few crucial factors such as choice of manifold representation, distance/similarity metrics and length of video sequence.

However, our motivation by the growing trends of these approaches begs the question: *How can we fuse together both exemplar and image set matching to further improve recognition ability?* Approaches that utilized both image sets and exemplars are few. Most notably, Wang et al. (2008) proposed a weighted distance measure involving both subspace and exemplar distances by means of local linear patches for recognition on the face manifold. While the fusion into a single distance measure is straightforward and efficient, the optimum setting for the weight parameter is difficult to determine and does not fully leverage interactions at the cluster level.

In this paper, we introduce a new fusion approach that incorporates both cluster subspace and exemplar point similarities derived within local appearance-based clusters for face recognition in video. In terms of the dual-feature representation, the learned cluster subspaces are capable of characterizing set variations at the cluster level while the selected exemplars encode detailed appearance information at the point (image) level. The subjects in video are recognized using a Bayesian *maximum-a posteriori* classifier, where a joint probability function that captures relevant dependencies is formulated using similarity metrics. The promising results obtained from an extensive evaluation on three face video databases – Honda/UCSD, CMU MoBo and ChokePoint, with further comparisons with other related methods have demonstrated the effectiveness of the proposed approach.

The rest of this paper is organized as follows. In Section 2, we briefly describe the problem setting and representations used. Section 3 elaborates on the proposed approach in detail. Experimental results and further analysis are discussed in Section 4. Section 5 concludes this paper.

2. Feature representation

In this work, we adopt a similar setting to that used in exemplar-based approaches, where face images in a video sequence are grouped into local appearance-based clusters in the initial step. These local clusters offer rich variational features that can be exploited. Intuitively, subspaces or manifolds describe coarse set-based variations while selected points within a cluster reflect fine appearance variations. In both training and test sequences, each subject class is represented in dual-feature mode – as subspaces (entire clusters) and points (exemplar images).

2.1. Problem formulation

Generally, we define a sequence of face images extracted from a video of subject c as an array of observed data in $\mathbb{R}^D$,

$$\mathbf{X}_c = \{\mathbf{x}_{c,1}, \mathbf{x}_{c,2}, \ldots, \mathbf{x}_{c,N_c}\}, \tag{1}$$

where N_c is the number of face images in the video, with the subject label of a C-class problem, $c \in \{1, 2, \ldots, C\}$ and each video is assumed to contain faces of the same person.

For each training video,[1] M number of clusters are extracted for each subject,

$$\mathbf{Z}_c = \{\mathbf{z}_{c,1}, \mathbf{z}_{c,2}, \ldots, \mathbf{z}_{c,M}\} \text{ , where} \tag{2}$$

$$\mathbf{z}_{c,m} = \{\mathbf{x}_{c,1}, \mathbf{x}_{c,2}, \ldots, \mathbf{x}_{c,N_m}\} \tag{3}$$

is the mth cluster, a vector of N_m images. Subsequently, an exemplar image is selected from each cluster, resulting in an exemplar set for the cth subject,

$$\mathbf{E}_c = \{\mathbf{e}_{c,1}, \mathbf{e}_{c,2}, \ldots, \mathbf{e}_{c,M}\} \tag{4}$$

where $\mathbf{e}_{c,m} \subseteq \mathbf{z}_{c,m}$.

For each test video $\mathbf{X}'$, each face image $\mathbf{x}'_i$ (similar as denoted in (1)) is a point feature of the unobserved data. Clusters are derived by partitioning the test video sequence into temporally continuous segments of length L,

$$\mathbf{z}'_k = \left\{\mathbf{x}'_{k,1}, \mathbf{x}'_{k,2}, \ldots, \mathbf{x}'_{k,L}\right\} \tag{5}$$

With that, the test cluster set for the cth class is denoted by,

$$\Theta_c = \{\boldsymbol{\theta}_{c,k,1}, \boldsymbol{\theta}_{c,k,2}, \ldots, \boldsymbol{\theta}_{c,k,N_c}\} \tag{6}$$

where the test cluster vector corresponding to the ith frame is assigned the kth cluster segment, $\boldsymbol{\theta}_{c,k,i} = \mathbf{z}'_k$ based on its associated image $\mathbf{x}'_i \in \mathbf{z}'_k$.

2.2. Cluster and exemplar extraction

With the large amount of face variations in each training video, the nonlinear dimensionality reduction method, Locally Linear Embedding (LLE) (Roweis and Saul, 2000) is applied to learn a low-dimensional embedding from its original data space. It is well known that LLE is capable of modeling the intrinsic structure of a nonlinear data manifold to capture meaningful embedding to better capture various face variations such as pose, expression and illumination.

Next, for each training video, the projected faces in LLE feature space are partitioned into clusters using the recently proposed spatio-temporal hierarchical agglomerative clustering (STHAC) algorithm (See and Eswaran, 2011), where consideration for both spatial and temporal distances tend to produce better clusters. The global spatio-temporal fusion scheme of STHAC is used in this work.

For each cluster, the face image nearest to cluster mean is selected as the exemplar. The final exemplar set consists of the extracted exemplars from all training videos, which can be succinctly summarized as

$$\mathbf{E} = \{e_{c,m} | c = 1, \ldots, C;\ m = 1, \ldots, M\} \tag{7}$$

2.3. Representation

For all cluster sets, features can be described using a subspace representation spanned by the images in the cluster where distance between subspaces or similarity measures can be derived. In our approach, each cluster $\mathbf{z}_i$ of the ith set is represented as a d-dimensional linear subspace by an orthonormal basis matrix

[1] Assumes one training video per subject. If more than one training video is used, image frames from all similar-class videos can be aggregated sequentially before extracting clusters.

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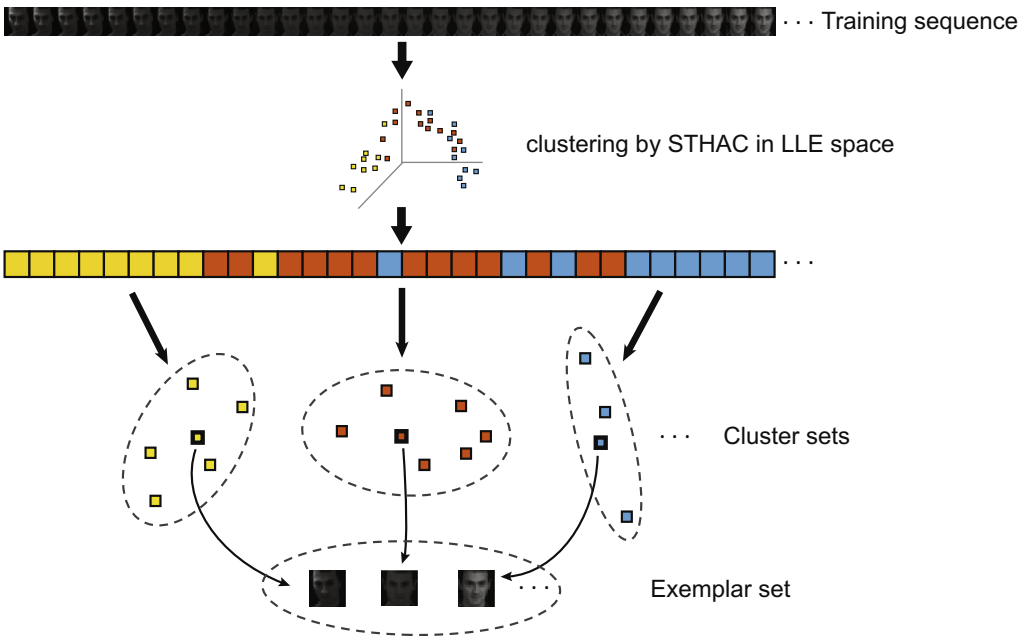

Fig. 1. Cluster-centric extraction of subspace and point features from a training video sequence. The cluster sets are characterized by linear subspaces while the exemplar set consists of exemplar points projected to NDMP feature space.

$\mathbf{P}_i \in \mathbb{R}^d$, obtained by the Singular Value Decomposition (SVD) of $\mathbf{z}_i\mathbf{z}_i^T = \mathbf{P}_i\tilde{\Lambda}\mathbf{P}_i^T$, where $\tilde{\Lambda}$ and $\mathbf{P}_i$ are the eigenvalue and eigenvector matrices of the d largest eigenvalues, respectively. Assume $\mathbf{P}_i \in \mathbb{R}^d$ and $\mathbf{P}_j \in \mathbb{R}^d$ denote the orthonormal bases for two subspaces $\mathbf{z}_i$ and $\mathbf{z}_j$, the SVD of $\mathbf{P}_i^T\mathbf{P}_j \in \mathbb{R}^{d\times d}$ is given as

$$\mathbf{P}_i^T\mathbf{P}_j = \mathbf{Q}_{ij}\Lambda\mathbf{Q}_{ji}^T \quad s.t.\ \Lambda = diag(\sigma_1, \ldots, \sigma_d) \tag{8}$$

where the singular values $\sigma_1, \ldots \sigma_d$ are the cosines of principal angles, also commonly known as canonical correlations. This forms the basic workings of the Mutual Subspace Method (MSM) where the cosine of the smallest principal angle (or largest canonical correlation) is used as the similarity measure between two subspaces. Other kernelized (Wolf and Shashua, 2003) and discriminant (Kim et al., 2007) variants are also used in the comparative experiments presented in Section 4.1.

For the training exemplar set, we employ the recently proposed nonlinear dimensionality reduction method called Neighborhood Discriminative Manifold Projection (NDMP) (See and Ahmad Fauzi, 2011). In brief, NDMP seeks to learn an optimal low-dimensional projection by considering both intra-class and inter-class reconstruction weights. Global structural and local neighborhood constraints are imposed within a constrained optimization problem, which can be solved as the generalized eigenvalue problem:

$$(\mathbf{E}\mathbf{M}_{intra}\mathbf{E}^T)\mathbf{A} = \lambda(\mathbf{E}\mathbf{M}_{inter}\mathbf{E}^T + \mathbf{E}\mathbf{E}^T)\mathbf{A} \tag{9}$$

where training exemplar set $\mathbf{E} \in \mathbb{R}^D$, and $\mathbf{M}_{intra}$ and $\mathbf{M}_{inter}$ are the intra-class and inter-class orthogonal weight matrices respectively. A new test sample $\mathbf{X}'$ can be projected to embedded space $\mathbb{R}^{\tilde{d}}$, by the linear transformation $\mathbf{Y}' = \mathbf{A}^T\mathbf{X}'$ where $\tilde{d} \ll D$. Detailed theoretical formulation of NDMP can be found in the work by See and Ahmad Fauzi (2011).

Fig. 1 summarizes the extraction of subspace and point features from a training video sequence through local class-specific appearance-based clusters.

3. Fusion approach to recognition

3.1. Dual-feature Bayesian classifier

The proposed recognition approach is formulated based on the evaluation of class hypotheses set $\mathbf{H} = \{H_c\}, c = 1 \ldots C$, by Bayesian classification, where the subject identity of a new sequence $\mathbf{X}$ can be found by estimating the *maximum-a posteriori* (MAP) decision rule[2]:

$$c^* = arg\ \max_C\ p(c|\mathbf{X}) \tag{10}$$

In general, assuming conditional independence between all observations, i.e. $\mathbf{x}_i \perp\!\!\!\perp \mathbf{x}_j |\ c$ where $i \neq j$, the posterior distribution over the class hypotheses at time frame N at the end of test sequence $\mathbf{X}$ can be expressed by Bayes rule,

$$p(c|\mathbf{X}) \equiv p(c|\mathbf{x}_1, \ldots, \mathbf{x}_N) \propto p(c)\prod_{i=1}^{N} p(\mathbf{x}_i|c) \tag{11}$$

This is the commonly known Naive Bayes classifier.

To further encode both subspace and point features, a Bayesian classifier can be modeled in the form of joint probability functions. In our earlier exemplar-driven approach (See et al., 2011), the joint probability function involved only the use of exemplar points and the causal relationship between exemplars and their respective classes,

$$p(c, \mathbf{E}, \mathbf{X}) = p(\mathbf{X}|c, \mathbf{E})p(\mathbf{E}|c)p(c) \tag{12}$$

From that, a new joint probability function is now proposed to fuse together relevant similarities within local clusters at both point and subspace levels,

$$p(c, \mathbf{E}, \mathbf{Z}, \Theta, \mathbf{X}) = p(\mathbf{X}|\mathbf{E}, \Theta)p(\Theta|\mathbf{Z})p(\mathbf{Z}|c)p(\mathbf{E}|c)p(c). \tag{13}$$

[2] For clarity purpose, we abuse the notation by simply denoting the test sequence as $\mathbf{X}$, contrary to earlier notation of $\mathbf{X}'$ in Section 2.3.

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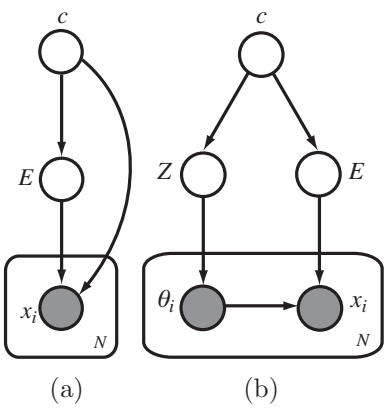

Fig. 2. Graphical models for (a) exemplar-driven, and (b) dual-feature Bayesian classifiers.

The conditional dependencies among the point feature $(\mathbf{E}, \mathbf{X})$ and subspace feature $(\mathbf{Z}, \Theta)$ variables are encoded in the Bayesian framework. Graphical models depicting both the exemplar-driven approach and the new approach are shown in Fig. 2.

To determine the class hypothesis at the end of a test video sequence, the posterior probability to be maximized is redefined as,

$$\begin{aligned} p(c|\mathbf{E},\mathbf{Z},\Theta,\mathbf{X}) &= \frac{p(\mathbf{X}|\mathbf{E},\Theta)p(\Theta|\mathbf{Z})p(\mathbf{Z}|c)p(\mathbf{E}|c)p(c)}{\sum_{c'} p(\mathbf{X}|\mathbf{E},\Theta)p(\Theta|\mathbf{Z})p(\mathbf{Z}|c')p(\mathbf{E}|')p(c')} \\ &\propto p(c)\prod_{i=1}^{N} p(\mathbf{x}_i|\mathbf{E},\Theta)p(\Theta|\mathbf{Z})p(\mathbf{Z}|c)p(\mathbf{E}|c) \\ &\propto p(c)\prod_{i=1}^{N} p(\mathbf{x}_i|\mathbf{E})p(\mathbf{x}_i|\Theta)p(\Theta|\mathbf{Z})p(\mathbf{Z}|c)p(\mathbf{E}|c) \end{aligned} \tag{14}$$

with the assumption of conditional independence between observations in $\mathbf{X}$ and law of total probability. Random variables Θ and $\mathbf{E}$ are also assumed independent of each other as they both represent different features types and have no mutual causality for classification.

Two additional assumptions are asserted to further simplify it to the final compact term. With the use of local clusters, each class is now represented by M number of clusters and exemplars (see Eqs. (2) and (3)). Hence, conditional probabilities involving Θ and $\mathbf{E}$ can be aggregated by sum rule for consistency in our framework. Finally, $p(\mathbf{x}_i|\Theta)$ and $p(\mathbf{Z}|c)$ are assumed non-informative terms and treated as constants as they intuitively insignificant to the decision rule. With that, we obtain as follows:

$$\begin{aligned} p(c|\mathbf{E},\mathbf{Z},\Theta,\mathbf{X}) &\equiv p(c|\mathbf{e}_{c,j},\mathbf{z}_{c,j},\boldsymbol{\theta}_i,\mathbf{x}_i) \\ &\propto p(c)\prod_{i=1}^{N} p(\mathbf{x}_i|\boldsymbol{\theta}_i)\sum_{j=1}^{M} p(\boldsymbol{\theta}_i|\mathbf{z}_{c,j})p(\mathbf{x}_i|\mathbf{e}_{c,j})p(\mathbf{e}_{c,j}|c)p(\mathbf{z}_{c,j}|c) \\ &\propto p(c)\prod_{i=1}^{N}\sum_{j=1}^{M} p(\boldsymbol{\theta}_i|\mathbf{z}_{c,j})p(\mathbf{x}_i|\mathbf{e}_{c,j})p(\mathbf{e}_{c,j}|c) \end{aligned} \tag{15}$$

Due to the limited sample size in our problem setting, good estimations of distribution can be challenging and easily result in overfitting or under-fitting of data. In this work, we exploit relevant similarity metrics involving point and subspace features derived from within the local clusters (see Fig. 3 for illustration). Concretely, these similarity metrics are computationally inexpensive and structurally intuitive for formulating the relevant probabilities. The likelihoods $p(\boldsymbol{\theta}_i|\mathbf{z}_{c,j})$ and $p(\mathbf{x}_i|\mathbf{e}_{c,j})$ reflect the probabilities of test point $\mathbf{x}$ and subspace $\boldsymbol{\theta}$ at frame i given their respective trained image clusters and exemplars. The conditional probability $p(\mathbf{e}_{c,j}|c)$ weigh the importance of each exemplar with respect to its parent class. Since we have no information on the correctness of hypothesis at the start of the observation sequence, class priors $p(c)$ are uniformly distributed over all the class hypotheses.

3.2. Cluster likelihood $p(\boldsymbol{\theta}_i|\mathbf{z}_{c,j})$

Subspace features extracted from both training and test cluster samples can be utilized to determine similarity between clusters in the form of principal angles or canonical correlations (as briefly discussed in Section 2.3).

The cluster likelihood, or likelihood of the observed cluster subspace $\boldsymbol{\theta}_i$ (corresponding to face image $\mathbf{x}_i$) given the training cluster subspace $\mathbf{z}$ is as follows,

$$p(\boldsymbol{\theta}_i|\mathbf{z}_{c,j}) = (1-\alpha)S_i^{CL}(\boldsymbol{\theta}_i,\mathbf{z}_{c,j}) + \alpha \tag{16}$$

where the normalized subspace (cluster) similarity metric is defined by the average of first r canonical correlations, $S_i^{CL}(\boldsymbol{\theta}_i,\mathbf{z}_{c,j}) = \sum_{l=1}^{r}\sigma_l/r$ where $r < d$. In our experiments, the value of r is fixed to obtain consistent correlation values. Parameter α is the lower bound of the similarity metric (asserting value range $[\alpha, 1.0]$), defining its degree of sensitivity.

3.3. Exemplar likelihood $p(\mathbf{x}_i|\mathbf{e}_{c,j})$

A point similarity metric is defined as the inverse squared Mahalanobis distance between the observed face image $\mathbf{x}_i$ and the jth exemplar of class c in NDMP-projected embedded space, $S_i^{EL}(\mathbf{x}_i,\mathbf{e}_{c,j}) = 1/(\mathbf{x}_i-\mathbf{e}_{c,j})\Sigma^{-1}(\mathbf{x}_i-\mathbf{e}_{c,j})^T$ where Σ is the common covariance matrix for all class samples (inclusive of the test sample). The similarities are sum-normalized across all classes. The choice of Mahalanobis metric here is assumed to intuitively embody well-behaved multinormal sample distributions with a common covariance matrix.

The exemplar likelihood, or likelihood of the observed face image $\mathbf{x}_i$ given the training exemplars $\mathbf{e}$ is formulated under stochastic selection rule as,

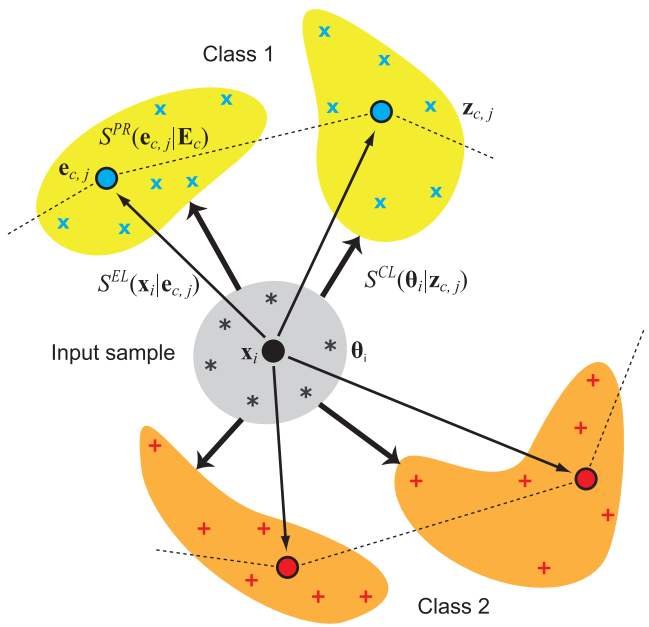

Fig. 3. Graphical illustration of similarity metrics involving exemplar points *(circles)* and cluster subspaces *(colored regions)* within local class-specific clusters.

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Fig. 4. Selected exemplars of two subjects from CMU Mobo *(top two rows)*, Honda/UCSD *(bottom two rows)* and ChokePoint *(middle two rows)* datasets.

Table 1
Recognition rates (%) of the evaluated methods on three datasets and their average across all datasets.

	CMU MoBo	Honda/UCSD	ChokePoint	Average
MajVote-PCA	83.5	60.8	71.8	72.03
MajVote-LDA	93.6	64.5	87.1	81.73
MajVote-NDMP	94.7	68.7	92.7	85.37
ProbVote	95.8	70.8	89.9	85.50
MSM	84.4	63.7	67.5	71.87
KPA	69.6	77.8	68.2	71.87
DCC-RandomSplit	81.4	62.7	72.8	72.30
DCC-Clusters	86.0	74.0	86.7	82.23
NB	95.3	75.9	91.7	87.63
EDB	96.1	77.6	92.1	88.60
MMD	95.3	78.4	92.2	88.63
DFB	97.2	85.9	93.6	92.23

$$p(\mathbf{x}_i|\mathbf{e}_{c,j}) = \frac{S_i^{EL}(\mathbf{x}_i, \mathbf{e}_{c,j})}{\sum_{k=1}^{C}\sum_{j=1}^{M} S_i^{EL}(\mathbf{x}_i, \mathbf{e}_{c,j})} \tag{17}$$

3.4. Exemplar prominence $p(\mathbf{e}_{c,j}|c)$

Causal relationship between exemplars and their respective parent classes can be represented by the conditional probability $p(\mathbf{e}_{c,j}|c)$. Intuitively, these conditional probabilities act as exemplar prominence weights for the exemplar likelihoods $p(\mathbf{x}_i|\mathbf{e}_{c,j})$, or coefficients of influence of the exemplar within its own class subspace. A point-to-subspace similarity metric is defined by the inverse ℓ^2-Hausdorff distance from each exemplar $\mathbf{e}_{c,j}$ to its corresponding class-specific exemplar subspace $\mathbf{E}_c$ in NDMP space, $S_{c,j}^{PR}(\mathbf{e}_{c,j}, \mathbf{E}_c) = 1/(\min_{\mathbf{e}' \in \mathbf{E}_c} \|\mathbf{e}_{c,j} - \mathbf{e}'\|)$.

The M similarities of each cth class are normalized by min–max normalization which is a linear mapping to $[0, 1]$. Hence, the exemplar prominence probability is formulated as,

$$p(\mathbf{e}_{c,j}|c) = \frac{S_{c,j}^{PR}(\mathbf{e}_{c,j}, \mathbf{E}_c)}{\sum_{j=1}^{M} S_{c,j}^{PR}(\mathbf{e}_{c,j}, \mathbf{E}_c)} \tag{18}$$

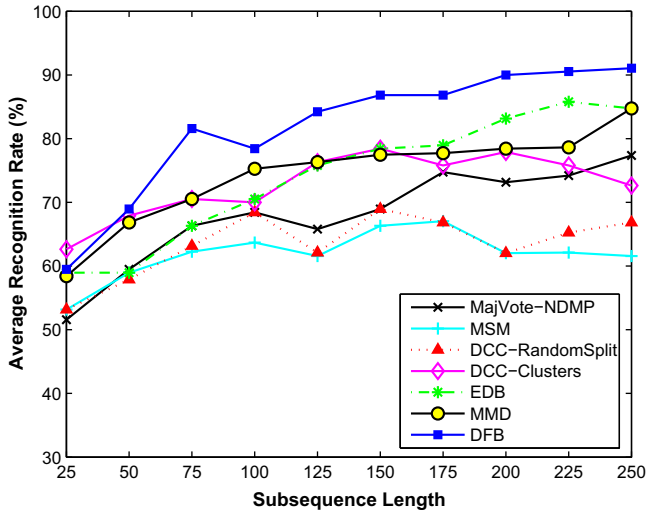

Fig. 5. Results of selected methods using different test subsequence lengths on the Honda/UCSD dataset.

This term can be pre-computed during training since it does not depend on observation sample **X**.

4. Experiments

The proposed method is evaluated in a video-based recognition framework on three public face video datasets. For each video sequence, faces are extracted using the Viola and Jones (2001) cascaded face detector and resampled to 32×32 pixel grayscale images. To cope with demanding head poses and rotations in the datasets, the undetected faces are re-tracked and localized using the robust IVT tracker (Ross et al., 2008) to ensure that the experiments are thoroughly evaluated with all possible views considered.[3] The face images are histogram equalized and no other pre-processing tasks were applied prior to recognition.

Based on accepted practice in most works, one video sequence is used for training and the remaining sequences are used for testing for each subject class. We formulate an extensive evaluation procedure by creating an augmented test set to introduce a variety of sequences with different starting frame positions and sequence lengths. The augmented test set is constructed by randomly sampling W subsequences of T different lengths from the original test videos belonging to each subject class, for all C subject classes. Thus, a total of $W \times T \times C$ subsequences in the augmented test set were sampled for our experiments. This evaluation procedure is designed to reduce video length bias and to better mimic realistic scenarios with arbitrary sets of views.

The first dataset we used is the CMU MoBo (Gross and Shi, 2001), one of the earliest datasets used for video-based face recognition, adapted from its original use for human identification from distance. It consists of 96 sequences of 24 different people walking on a treadmill (4 videos per person). Each video contains about 300 frames. The second dataset, Honda/UCSD (Lee et al., 2005) was collected specifically for face recognition in video and is one of the more challenging public dataset. We consider their first subset, which has 59 sequences of 20 different people (each person has at least 2 videos). Each video contains about

[3] The implementations of some works (Wang et al., 2008) directly obtain only detected faces while ignoring undetected faces as outliers, excluding them from the final image set.

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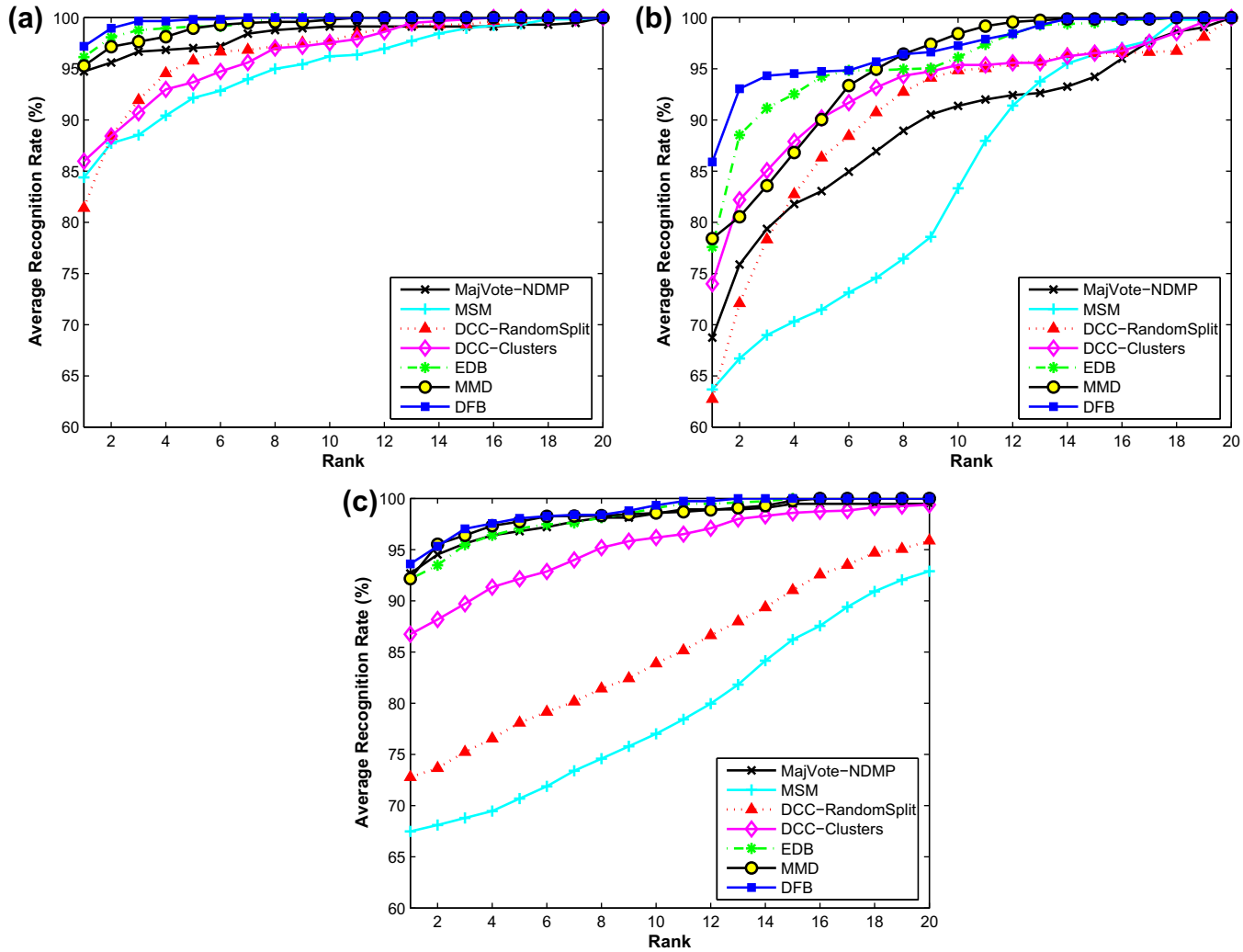

Fig. 6. Rank-based evaluation of selected methods on the (a) CMU MoBo, (b) Honda/UCSD, and (c) ChokePoint datasets.

300–600 frames, comprising of large pose and expression variations with significant out-of-plane head rotations. The third dataset is the recently collected ChokePoint dataset (Wong et al., 2011) by NICTA for person identification under real-world surveillance conditions. We consider their first subset (Portal 1), which has 600 sequences of 25 different people. Each person has 24 different sequences, comprising of a combination of 4 sequence shots, 3 camera angles, and 2 movement modes (entering and leaving portal). The cropped face images provided with the dataset contain variations in terms of illumination conditions and pose, video quality, as well as misalignments due to the presence of slight occlusions. To simplify our experimental set up for this dataset, we only sample sequences taken by Camera 1 (single camera) from the leaving portal mode.

To generate the augmented test set, we use the following sampling parameters, in (W, T, C): CMU MoBo (12, 5, 20), Honda/UCSD (10, 5, 20) and ChokePoint (9, 5, 25).[4] The number of subsequence lengths is fixed at $T = 5$ to provide fair and consistent evaluation across all tested datasets. The subsequence lengths are determined by T equally distributed intervals of the original sequence length. Meanwhile, W is arbitrarily decided on the basis of creating sufficient samples for each tested class.

[4] Details of the protocol for generating the augmented test set can be found at http://pesona.mmu.edu.my/johnsee/research/vfr.

4.1. Comparative evaluation

In our experiments, we compared the performance of the proposed Dual-Feature Bayes (DFB) classifier to the following methods:

- Exemplar-based majority voting (MajVote) by Euclidean nearest neighbor (NN) distance in PCA, LDA and NDMP projected spaces, with vote taken at each frame
- Exemplar-based probabilistic voting (ProbVote) in NDMP projected space, where the NN distances are normalized to probabilities and aggregated cumulatively by sum rule, with vote taken at each frame
- Mutual subspace method (MSM)
- Kernel principal angles (KPA)
- Discriminative canonical correlations (DCC), with training for each class conducted by (i) randomly partitioning into two class-specific sets, and (ii) partitioning into M class-specific cluster sets as extracted by STHAC
- Manifold–manifold distance (MMD) method
- Bayesian MAP classifiers – Naive Bayes (NB) and Exemplar-Driven Bayes (EDB)

The MajVote, ProbVote, NB and EDB are exemplar-based methods, MSM, KPA and DCC are image set-based methods, while MMD

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and DFB, our proposed approach, are a feature-wise fusion of both types of schemes.

The optimal parameters for each approach were determined by experiments. For the voting methods, PCA features are set to retain 95% of data energy, while for both LDA and NDMP features, the dimension is set to the number of classes minus one. The exemplar training set in all Bayesian classifiers (NB/EDB/DFB) are projected to NDMP space, with dimension similarly set to the number of classes minus one. The NDMP dimensionality reduction method is selected for its ability to extract meaningful discriminative features from the highly nonlinear training set manifold with complex sample variability (See and Ahmad Fauzi, 2011).

For methods that require clustering during training (Voting methods, DCC with clusters, NB, EDB, DFB), the number of clusters per subject was determined empirically (details can be found in the work by See and Eswaran (2011)), *i.e.* $M = 7$ for Honda/UCSD and $M = 6$ for CMU MoBo and ChokePoint. In Fig. 4, some sample exemplar sets extracted from training videos of the datasets show the variability of face appearances obtained using STHAC.

For MSM/KPA/DCC/DFB, the dimensionality of PCA subspaces learned for each image set is set to 10, representing about 98% of data energy of the set. The kernel function used in KPA is a Gaussian radial basis function (RBF) kernel, a promising choice suggested in the original authors' implementation (Wolf and Shashua, 2003). For MMD, the weighting parameter is set to 0.5 for equal weights, as suggested by Wang et al. (2008). Cluster subspaces for DFB are represented using DCC while the temporal segment length L and subspace similarity sensitivity α are set to good empirical values of $L = 20$ and $\alpha = 0.75$.

4.2. Results

The experimental results of the evaluated approaches (as listed in Section 4.1) for all three datasets in terms of average recognition rate, are presented in Table 1. As observed, the proposed DFB approach obtained the best performance among all evaluated approaches, for all three datasets. Its effectiveness is most prominent in the challenging Honda/UCSD dataset compared to other datasets, while it is also distinctly superior when results are averaged across all datasets (92.23%). The performance of the image set-based approaches were noticeably poor in the Honda/UCSD and ChokePoint datasets possibly due to their inadequacy in characterizing complex and rapidly changing head poses. Overall, approaches that employ the described Bayesian framework (NB, EDB, DFB) yielded better performance than conventional voting schemes. Also, the feature fusion approaches (MMD, DFB) seemed to produce promising results. In the case of DFB, the unification of features from both video frames (point) and video segments (subspace) using a temporally-driven classification framework, provide a holistic representation for improved recogniton.

4.3. Further discussions

On closer examination of different video subsequence lengths (using $T = 10$), the DFB approach gradually outperformed the other approaches as the subsequence length increases, as shown in Fig. 5 on the Honda/UCSD dataset. This is a strong characteristic of our Bayesian framework where temporally accummulated posteriors can increase the accuracy of converging towards a candidate subject. On the other hand, the performance of image set-based approaches appeared highly erratic and they are not able to achieve recognition rates as high as the Bayesian methods. Since the variability of faces increase in longer videos, characterizing entire videos as image set subspaces does not necessarily ensure that finer appearance information are well captured. Likewise, exemplar-based methods suffer from the rigidity of frame-wise classification (more so using crude voting strategies), ignoring the usefulness of set representations at a coarser granularity. For a general summary of the one-to-many identification performance, the cumulative match curves of the evaluated methods shown in Fig. 6 demonstrate the robustness of the proposed approach across rank.

Though it is possible to improve the overall recognition performance of the evaluated methods by applying state-of-the-art illumination normalization techniques in the pre-processing step (such as one proposed by Tan and Triggs (2010)), we abide by the widely accepted practice of normalizing pixel intensities by histogram equalization across all frames in a video sequence, which is common in many notable works in literature (Arandjelovic et al., 2005; Wang et al., 2008; Harandi et al., 2011). Likewise, this work addresses the classification aspect of the VFR problem while fixing to constant the methods applied to all other steps.

Briefly on the computational cost, we report an average classification computation time of 0.103 s per sequence (average sequence length of 150 frames) for the DFB on the Honda/UCSD (10, 5, 20) using a Pentium IV 2.8 GHz, 1.5 Gb RAM machine. In comparison to the usage of dual features, single feature methods obtained mixed performances *i.e.* exemplar-feature-only methods – NB: (0.078 s), EDB: (0.093 s), and image-set-feature-only methods – MSM (0.163 s), DCC-RandomSplit (0.236 s), DCC-Clusters (0.755 s). The DFB also fared better than the MMD (0.256 s) in terms of classification speed. This is strong indication that the proposed Bayesian framework employed for recognition remains a key factor behind its efficiency. The calculation of the similarity metrics involved is inexpensive and does not incur much computational overhead, with the exception of the cluster likelihood, which can be pre-computed prior to matching. This burden of computation is passed to the training stage where the DFB takes 83.987 s, or about 10^2 times the duration needed to train the EDB (0.813 s), this in return for a 4% improvement in accuracy. DFB is also only marginally longer than the training time required for DCC-Clusters approach (77.328 s). Nonetheless, it is worth noting that efficiency in the classification step (in testing stage) is ultimately important for practical applications.

5. Conclusion

We propose a novel dual-feature classification approach to face recognition in video that fuses an essential set of similarities derived within local appearance-based clusters. Both subspace and point features characterize a diverse set of representations, whereby the learned subspaces capture intrinsic variations at the subspace level while the selected exemplars encode the appearance information of the trained subject at the point level. Within the Bayesian *maximum-a posteriori* (MAP) classification framework, a joint probability function is modeled using relevant similarity metrics to recognize the identity of subjects in video. In a comprehensive evaluation on three face video datasets – CMU MoBo, Honda/UCSD and Chokepoint, the proposed approach demonstrated promising results in comparison with conventional single feature approaches. Our work opens up future directions towards the application of a compact Bayesian similarity-based framework for various temporally-driven data, and also further exploration into different feature representations that can be leveraged.

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